

# Prosodic — Metric System Specification

Definitive specification of all computational metrics, craft signals, algorithmic thresholds, and scoring models across the Prosodic analysis engine suite.

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## 1. Architectural Framework & Normalization

All raw engine outputs are mapped onto a unified, comparable scale through `domain/final_result_converter.py`, ensuring multi-engine blends do not distort or invert relative weights.

- **Standard Scale Output Range:** `0.0` to `1.0` (Continuous).
- **Standard Polarity:** Always `normal` (higher numerical value = stronger craft expression).
- **Clipping:** Clamped defensively to `[0.0, 1.0]`.

Source Metric Registry ( SCALES )

| Source Key                   | Raw Range   | Polarity | Originating Engine          | Description                                              |
|------------------------------|-------------|----------|-----------------------------|----------------------------------------------------------|
| phoneme_rhyme_score          | 0.0 – 1.0   | normal   | phoneme_engine.py           | Phonetic similarity between ARPABET nucleus & coda       |
| perceptual_family_score      | 0.0 – 1.0   | normal   | perceptual_family_engine.py | Confidence of vowel family membership                    |
| pattern_activity_score       | 0.0 – 1.0   | normal   | pattern_reader_engine.py    | Composite family activity (frequency, line span, stress) |
| semantic_similarity          | 0.0 – 1.0   | normal   | semantics_engine.py         | spaCy word-vector cosine distance                        |
| density_internal             | 0.0 – 100.0 | normal   | density_engine.py           | Percentage of syllables rhyming internally               |
| density_multisyllabic        | 0.0 – 100.0 | normal   | density_engine.py           | Percentage of syllables in compound rhyme blocks         |
| density_motif                | 0.0 – 100.0 | normal   | density_engine.py           | Motif family palette coverage per bar                    |
| suggestion_rhyme_score       | 0.0 – 100.0 | normal   | suggestion_engine.py        | Phoneme rhyme score expanded to 0-100                    |
| suggestion_thesaurus_score   | 0.0 – 100.0 | normal   | suggestion_engine.py        | Moby Thesaurus tiered relevance score                    |
| suggestion_syllable_priority | 0.0 – 2.0   | normal   | suggestion_engine.py        | Ordinal priority: 0 (off), 1 (±1 syl), 2 (exact)         |
| stress_signal_confidence     | 0.0 – 1.0   | normal   | stress_signals.py           | Confidence score for stress mismatch detection           |
| phrase_boundary_weight       | 1.5 – 3.0   | normal   | phrase_container_engine.py  | Combined weight of fired boundary signals                |

2. Rhyme & Phonetic Architecture

Core Thresholds & Classifications

- **Phoneme Rhyme Score:** Continuous float in [0.0, 1.0] measuring phonetic alignment.
- **Rhyme Edge Threshold ( RHYME\_THRESHOLD = 0.78 ):** Minimum similarity score required to establish a valid rhyme edge in graph clustering.
- **Near-Rhyme Same Vowel Score ( NEAR\_RHYME\_SAME\_VOWEL\_SCORE = 0.75 ):** Baseline for near-rhymes sharing vowel base with different stress digit or rhotic coloring (e.g. ER1 vs ER2 ).

## Qualitative Rhyme Classifications ( `RhymeType` )

- `PERFECT` — Identical vowel nucleus and identical coda consonants (e.g., *mind* / *blind*).
- `SLANT` — Shared vowel nucleus with phonetically proximate coda consonants.
- `NEAR` — Shared vowel with minor coda divergence.
- `FAMILY` — Membership in the same perceptual vowel sound family.
- `NONE` — Disparate phonetics (<0.78).

## The 12 Canonical Perceptual Vowel Families

| Family Identifier       | ARPABET Nucleus | Anchor Words                      | Example Members                                |
|-------------------------|-----------------|-----------------------------------|------------------------------------------------|
| <code>R_FAMILY</code>   | <code>ER</code> | worst, curse, verse, first, birth | hurt, rehearsal, church, merch, turnt, diverse |
| <code>AY_FAMILY</code>  | <code>EY</code> | day, way, name, face, change      | flame, rain, train, late, state, claim         |
| <code>EE_FAMILY</code>  | <code>IY</code> | see, free, feel, real, deep       | street, dream, clean, scene, believe           |
| <code>OW_FAMILY</code>  | <code>OW</code> | know, flow, soul, cold, road      | home, phone, stone, throne, grown              |
| <code>AH_FAMILY</code>  | <code>AH</code> | blood, love, tough, run, trust    | young, done, dust, rush, club, sub             |
| <code>AY2_FAMILY</code> | <code>AY</code> | life, night, mind, time, light    | sky, high, drive, alive, desire, fire          |
| <code>OO_FAMILY</code>  | <code>UW</code> | through, true, move, cool, truth  | choose, rules, school, groove, mood            |
| <code>AW_FAMILY</code>  | <code>AW</code> | down, found, ground, sound, out   | crowd, loud, town, power, cloud                |
| <code>AE_FAMILY</code>  | <code>AE</code> | back, track, rap, trap, black     | stand, brand, plan, slam, exam                 |
| <code>OH_FAMILY</code>  | <code>AO</code> | talk, call, fall, thought, law    | raw, cause, wall, long, strong, wrong          |
| <code>IH_FAMILY</code>  | <code>IH</code> | win, begin, spin, king, think     | skin, bring, distinct, click, stick            |
| <code>EH_FAMILY</code>  | <code>EH</code> | head, dead, breath, best, death   | chest, blessed, stressed, check, respect       |

## AAVE Phonological Transformations ( `aave_phonology.py` )

1. **Consonant Cluster Reduction:** Word-final stop deleted in homorganic clusters (e.g. `/P AE1 S T/` → `/P AE1 S/` allowing *past* to rhyme with *class*; *hand* → *man*).
2. **Monophthongization:** Glide weakening of `/AY/` → `/AA/` (e.g. `/T AY1 M/` → `/T AA1 M/` allowing *time* to rhyme with *calm*; *mine* → *man*).
3. **Postvocalic R-Vocalization:** Deletion of rhotic codas (e.g. `/M A01 R/` → `/M A01/` rhyming with *saw*; `/ER/` → `/AH/` merging *bird* and *bud*).

### 3. Syllable & Rhyme Density

#### Density Component Formulas

Internal Rhyme Density (%) = (Highlighted Rhyme Syllables / Total Syllables) × 100  
 Multisyllabic Density (%) = (Deduplicated Compound Syllables / Total Syllables) × 100  
 Motif Recurrence Density (%) = (Distinct Colors in Bar / Total Palette in Verse) × 100

#### Syllabic Load & Pocketing Control

- **Syllables Per Beat (SPB):** Syllable count distributed across quarter-note pulses.
- **Density Control Window:** Optimal hip-hop pocket sits in `2.5 - 4.5 SPB`.

#### Density Gradient Arc Shapes ( `ArcType` )

- `BUILDING` — Syllabic load rises steadily toward the close of a section.
- `RELEASING` — Syllabic load descends from high density to provide breath and resolution.
- `FLAT` — Variance across bars is <20% ( `FLAT_VARIANCE_RATIO = 0.20` ).
- `PEAK_AND_RELEASE` — Sustained high-density peak (≥25% duration) followed by open release.
- `COMPLEX` — Undulating or multi-peak density profile.

### 4. Cadence, Pocket & Beat Grid Mechanics

#### The 16-Step Beat Grid

- **Resolution ( `GRID_SIZE = 16` ):** 16 sixteenth-note slots per 4/4 bar.
- **Strong Downbeats:** Positions `0, 4, 8, 12` (Beats 1, 2, 3, 4).
- **The Pocket:** Positions `4` and `12` (Snare downbeats on Beats 2 and 4).
- **Tolerances:** `POCKET_WINDOW = 1` (±1 sixteenth note); `NUDGE_WINDOW = 2`.

#### Flow Signatures

- `On-Grid` — >75% of stressed syllables land on strong downbeats (0, 4, 8, 12).
- `Syncopated` — >50% of stressed syllables land on the snare pocket (4, 12).
- `Floating` — <35% on-beat alignment; decoupled from the rigid pulse.
- `Pocket Jumper` — Dynamic alternation between on-beat landing and syncopated runs.

Cadence Signal Taxonomy (Stress Mismatches)

| Signal Identifier      | Trigger Condition                                                                     | Craft Meaning                                    |
|------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------|
| promotion              | Lexically weak syllable (0) lands on stressed grid position ( $\geq 1$ ).             | Performed rhythmic emphasis on structural word.  |
| demotion               | Lexically stressed syllable (1, 2) lands off-beat within <code>POCKET_WINDOW</code> . | Softened delivery adjacent to the downbeat.      |
| syncopation            | Lexically stressed syllable lands off-beat beyond <code>POCKET_WINDOW</code> .        | Deliberate rhythmic displacement between pulses. |
| trochaic_inversion     | Rising grid prominence over falling lexical stress.                                   | Metrical reversal creating expressive tension.   |
| stress_clash           | Adjacent lexically stressed syllables colliding without unstressed buffer.            | Trigger environment for English's Rhythm Rule.   |
| stress_lapse           | $\geq 3$ consecutive unstressed syllables ( <code>LAPSE_RUN_LENGTH = 3</code> ).      | Rhythmic lull or rapid syllabic patter.          |
| secondary_recruitment  | Secondary stress (2) lands on strong beat while primary (1) does not.                 | Anchoring rhythm on secondary word syllable.     |
| level_stress_ambiguity | Word has multiple dictionary pronunciations with contested stress.                    | Lexical ambiguity in CMU dictionary.             |

Deliberateness Gate

- `likely_automatic` — Occurs within a `stress_clash` context (automatic Rhythm Rule retraction).
- `possible_deliberate` — Word recurs on  $\geq 2$  other lines without clash context ( `RECURRENCE_MIN = 2` ).
- `uncertain` — Standard baseline; under-claims intent to avoid false assertions.

## 5. Compositional Structure & Phrase Packaging

### Boundary Detection Rules ( `phrase_container_engine.py` )

A structural container break is declared when cumulative signal weight exceeds `BOUNDARY_THRESHOLD = 1.5` and at least `MIN_SIGNALS_FIRED = 2` individual signals fire:

| Signal Name                    | Weight | Threshold Condition                                                                                                |
|--------------------------------|--------|--------------------------------------------------------------------------------------------------------------------|
| <code>rhyme_resolution</code>  | 0.8    | Terminal rhyme scheme completes and resolves on this bar.                                                          |
| <code>rest_bar</code>          | 0.7    | Line has $\leq 3$ words ( <code>REST_BAR_MAX_WORDS = 3</code> ).                                                   |
| <code>density_drop</code>      | 0.6    | Prior line internal density $\geq 40\%$ and current line drops $> 40\%$ ( <code>DENSITY_DROP_RATIO = 0.6</code> ). |
| <code>syllable_reset</code>    | 0.5    | Line syllable count drops to $< 60\%$ of prior line ( <code>SYLLABLE_RESET_RATIO = 0.6</code> ).                   |
| <code>line_length_shift</code> | 0.4    | Line word count differs by $> 4$ words from previous ( <code>LINE_LENGTH_SHIFT_WORD_DIFF = 4</code> ).             |

## 6. Visual Motifs & Pattern Reading

### Perceptual Family Activity Score Formula

$$\text{Activity Score} = (\text{Member Count Norm} \times 0.40) + (\text{Line Spread} \times 0.35) + (\text{Stress Ratio} \times 0.25)$$

Where:

- Member Count Norm =  $\min(\text{Members}, 10) / 10$
- Line Spread = Unique Lines with Family / Total Lines
- Stress Ratio = Stressed Members / Total Members

- **Active Family Threshold:** `Activity Score  $\geq 0.25$`  confirms intentional deployment.
- **Parallel Pocketing:** Flagged when top 2 active families have activity scores within 15% of each other.
- **Motif Cluster Strength ( `cluster_strength` ):** Continuous float in `[0.0, 1.0]` measuring semantic tightness around an anchor word.

## 7. Semantics, Concreteness & Suggestions

### Lexical & Semantic Dimensions

- **Semantic Similarity ( `semantic_similarity` ):** Cosine distance in `[0.0, 1.0]` using spaCy `en_core_web_md` vectors.
- **Brysbaert Concreteness Score:** Scale of `1.0 - 5.0` (1.0 = abstract concepts like "freedom"; 5.0 = tangible objects like "brick").

## Suggestion Composite Ranking Score ( `suggestion_engine.py` )

Composite Score = (Rhyme Score × 0.55) + (Thesaurus Score × 0.30) + (Syllable Priority × 0.15)

- **Rhyme Score (55%):** Raw phonetic similarity scaled to 0–100.
- **Thesaurus Score (30%):** Moby Thesaurus depth tiers: `0, 30-70, 85, 90, 100` .
- **Syllable Priority (15%):** 0 = off-target, 1 = ±1 syllable, 2 = exact syllable match.

## 8. Literary & Rhetorical Device Detection

Evaluated across 12 core rhetorical detectors in `device_detection_engine.py`. Frequency rates are computed as events per 100 lines.

### Status Tiers

- `untapped` — Below emerging threshold ( $< em\_threshold$ ).
- `emerging` — Present occasionally ( $\geq em\_threshold$ ).
- `signature` — Frequently deployed; core artistic fingerprint ( $\geq sig\_threshold$ ).
- `overused` — Exceeds natural balance ( $> overuse\_threshold$ ).

### The 12 Detectors & Status Thresholds

| Device              | Category   | Detection Heuristic                                                                     | Emerging | Signature | Overuse |
|---------------------|------------|-----------------------------------------------------------------------------------------|----------|-----------|---------|
| Alliteration        | Sound      | $\geq 2$ consecutive non-stop words sharing initial consonant sound.                    | 5.0      | 18.0      | 50.0    |
| Assonance           | Sound      | $\geq 3$ non-stop words in a line sharing stressed vowel sound.                         | 4.0      | 15.0      | 45.0    |
| Anaphora            | Structural | Same word opens $\geq 2$ consecutive lines ( $\geq 3$ chars, non-stop).                 | 2.0      | 8.0       | 30.0    |
| Epistrophe          | Structural | Same word closes $\geq 2$ consecutive lines.                                            | 1.5      | 5.0       | 20.0    |
| Anadiplosis         | Structural | Last word of line N matches first word of line N+1.                                     | 1.0      | 4.0       | 15.0    |
| Simile              | Comparison | Direct comparison: "like a/an", "as ... as", "like the".                                | 3.0      | 10.0      | 35.0    |
| Rhetorical Question | Rhetorical | Line ending with a question mark ( <code>?</code> ).                                    | 1.5      | 6.0       | 20.0    |
| Repetition          | Emphasis   | Key content word ( $> 3$ chars) appears $\geq 4$ times in song.                         | 1.0      | 3.0       | 8.0     |
| Enumeration         | Structural | $\geq 2$ commas in a line (stacking 3+ listed items).                                   | 2.0      | 8.0       | 25.0    |
| Polysyndeton        | Structural | $\geq 3$ coordinating conjunctions ( <i>and</i> / <i>but</i> / <i>or</i> ) in one line. | 1.0      | 4.0       | 15.0    |
| Asyndeton           | Structural | $\geq 2$ commas in a line without coordinating conjunctions.                            | 1.5      | 6.0       | 20.0    |
| Parallelism         | Structural | Consecutive lines sharing identical 2-word opening syntax.                              | 1.5      | 6.0       | 20.0    |

## 9. Phonoaffective Signature Framework

Formulated in `aspiration_gap.py` to measure emotional encoding and acoustic intentionality.

### The 8-Component Formula

$$E = (PT \times w1) + (PP \times w2) + (SF \times w3) + (CM \times w4) + (IS \times w5) + (TP \times w6) + (SA \times w7) + (SD \times w8)$$

- **PT (Phonemic Texture):** Ratio of plosives/hard consonants vs liquids/fricatives, vowel openness.
- **PP (Prosodic Pressure):** Syllable density, delivery compression, stress inversion rate.
- **SF (Semantic Field Gravity):** Semantic valence, imagery clustering, thematic cohesion.
- **CM (Cultural Memory):** Vernacular frequency, regional marker density, AAVE bridges.
- **IS (Intentionality Signal):** Revision count, dictionary override telemetry.
- **TP (Temporal Position):** Section trajectory (Verse 1 setup vs Climax vs Outro).
- **SA (Sonic Aspiration):** Selected emotional target from the 15 canonical labels (*aggressive, celebratory, contemplative, defiant, desperate, grieving, hopeful, hungry, melancholic, nostalgic, peaceful, proud, raw, triumphant, vulnerable*).
- **SD (Subtext Delta):** Discrepancy between surface semantic sentiment and acoustic tension.
- **Aspiration Gap Score ( `aspiration_gap_score` ):** Euclidean distance bounded in `[0.0, 1.0]` (0.0 = perfect alignment).

## 10. Longitudinal Mastery & Progress Tracking

Evaluated across catalog history in `mastery_engine.py`.

### Activation Gate (Unlock Thresholds)

- $\geq 3$  distinct songs analyzed
- $\geq 40$  total rhyme events recorded
- $\geq 20$  total cadence events recorded
- $\geq 15$  total bars analyzed

### Proficiency Tiers

- Developing — 0 to 39
- Approaching — 40 to 64
- Competent — 65 to 81
- Exemplary — 82 to 100

### The 6 Core Skill Category Scoring Models

| Category               | Sub-Metric Breakdown                                                  | Scoring Model                                                                                                |
|------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Flow & Cadence         | Density control in 2.5–4.5 SPB, Inversion rate, Cadence class variety | $\text{Density Control} \times 0.55 + \text{Inversion Score} \times 0.30 + \text{Variety Bonus} \times 0.15$ |
| Rhyme Architecture     | Average phonetic similarity, Rhyme type diversity, AAVE bridge %      | $\text{Avg Similarity} \times 80 + \text{Type Diversity Bonus (max 20)}$                                     |
| Internal Rhyme Density | Density gradient mean, Peak density %, Cross-bar consistency          | $\text{Avg Density} \times 70 + \text{Consistency} \times 0.30$                                              |
| Multisyllabic Flow     | Syllable compression rate %, High-density lines ( $>3.5$ SPB) %       | $\text{Compression Rate} \times 0.60 + \text{High Density Rate} \times 0.40$                                 |
| Motif Architecture     | Semantic cluster strength, Unique semantic field diversity            | $\text{Avg Cluster Strength} \times 75 + \text{Field Diversity Bonus (max 25)}$                              |
| Stress Architecture    | Inversion control (ideal 0.15–0.30), Stress pattern variety           | $\text{Inversion Control} \times 0.65 + \text{Pattern Diversity} \times 0.35$                                |

### Trend Tracking

- **Early Window:** Earliest 40% of catalog by timestamp.
- **Recent Window:** Latest 40% of catalog by timestamp.
- **Delta:**  $\Delta = \text{Score\_recent} - \text{Score\_early}$  ( $\geq +5$ : up,  $\leq -5$ : down, otherwise: neutral).